Sahaj Bhadja

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Summary

Software engineer with **C/C++, operating-system scheduling, concurrency, Python automation, debugging, testing, and cross-platform delivery**, interested in reliable embedded software, wireless connectivity, low-level systems, and IoT products.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built Python services with explicit state transitions, failure isolation, and input validation, processing 5,000+ monthly jobs while increasing concurrent throughput 3x and reliably sustaining 99%+ production reliability.
- Debugged performance and data-consistency issues through structured error handling, validation, reconciliation, and state tracking, consistently reducing database latency from 1.2 seconds to 350 milliseconds for 500+ users.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Implemented multithreaded Python processing using ThreadPoolExecutor, handling 1,000+ files per batch with secure ingestion, fault-tolerant execution, diagnostic logging, automated tests, and reliably stable performance under peak loads.
- Investigated OCR failures, model timeouts, and API instability across 10,000+ daily requests, implementing fallback paths and reproducible validation while reliably reducing end-to-end feedback turnaround time by 60%.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*

- Developed performance-sensitive Python and PyTorch components processing 100,000+ trajectory records, improving long-horizon planning 18% through simulation, debugging, reproducible testing, model optimization, and iterative feature integration at scale.
- Modeled time-sensitive interactions across ego, road, and nearby-agent signals, improving closed-loop stability 15% while analyzing edge cases, system behavior, implementation tradeoffs, and experimental results at production scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed cross-platform Java and Python microservices across Windows, macOS, and Linux, automating builds and 20+ unit and integration tests through Jenkins, PyTest, and Mockito at production scale.
- Debugged and optimized API components using object-oriented patterns, caching, and field filtering, improving response times 30%, reducing payload sizes 25%, and lowering technical debt 20% across release cycles.

Cyber Security and Data Mining: Detecting Malicious URL **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*

- Built a Python security-detection pipeline over 650,000 URLs, engineering 21 behavioral indicators for IP presence, suspicious keywords, and structural anomalies to identify phishing and malware accurately at scale.
- Benchmarked Random Forest, recurrent, and convolutional models, achieving 96%+ detection accuracy while reliably analyzing performance tradeoffs and generating reproducible insights from high-volume malicious and benign input patterns.

Skills

Languages: C, C++, Python, Java, SQL, Shell, Bash, JavaScript, TypeScript
Systems: Linux, Unix, operating systems, CPU scheduling, concurrency, multithreading, resource allocation
Development: OOP, data structures, algorithms, system design, REST APIs, caching, fault tolerance
Testing: Git, CI/CD, Jenkins, PyTest, Mockito, unit testing, integration testing, test automation
Analysis: Debugging, performance optimization, logging, validation, NumPy, Pandas, PyTorch, Matplotlib

Projects

CPU-scheduling Simulator Live | *C++, Operating Systems*

- Implemented six C++ scheduling algorithms with edge-case handling, automated waiting and turnaround metrics, and Gantt visualization, testing 1,000+ process combinations while reliably reducing average turnaround time 25%.

Comprehensive Analysis of Blockchain for User Authentication | *IoT Security Research*

- Explored secure IoT device authentication and role-based access control across four permission tiers and three user-access scenarios, designing decentralized verification workflows that reduced repeated key-verification steps by 50%.

CaptionCraft - Image Caption Generator Live | *Python, PyTorch, Model Analysis*

- Benchmarked BLIP, GIT, and ViT-GPT-2 transformer architectures across 10,000 images and 100,000 captions, improving accuracy from 81% to 96% while building scalable, real-time inference APIs for visual understanding.